Chapter 14 - Probability

#Probability

Probability is the measure of likelihood of an event, ranging from 0 (impossible) to 1 (certain).

#Some Basic Terms including Independent Cases

Experiment

The processes which produce different possible outcomes or cases, are known as the experiments.

The experiment which produces unpredictable results each time even under identical conditions, is known as the random experiment.

The set of all possible outcomes of a random experiment is called the sample space.

Trial and Event

Performing a random experiment is called a trial.

Outcome or a combination of outcomes of an experiment is termed as an event.

Exhaustive cases

The number of cases which include all possible outcomes of a random experiment is said to be the exhaustive cases for the experiment.

Equally likely cases

While performing experiments, if any one of the possible outcomes may occur but no one case can be expected to occur more than the other then the cases are said to be equally likely.

Mutually exclusive cases/events

Two or more events are said to be mutually exclusive if their simultaneous occurrence is not possible.

Favourable cases

The cases or the outcomes of a random experiment which entail the happening of an event, are known as the cases favourable to that event.

Independent cases/events

Two events are said to be independent if the occurrence of one event does not affect the occurrence of the other.

Permutation and Combination

Permutation	Combination
Permutation of a set of objects means the arrangement of objects in some order.	Combination of a set of objects means the selection of objects without regard to any order of arrangement.
The total number of permutations formed by arranging r objects out of n objects is $^{n}P_{r} = \frac{n!}{(n-r)!}, (r \le n)$	The total number of combinations formed by selecting r objects out of n objects is $^{n}C_{r} = \frac{n!}{r!\,(n-r!)}, (r \le n)$

#Mathematical Definition of Probability

If there are n exhaustive, mutually exclusive and equally likely cases and m of them are favourable to an event E then the probability of the happy of an event E is

$$P(E) = \frac{m}{n}$$

The probability $P(E)$ of happening of an event E satisfies the following property:

1. $0 \le P(E) \le 1$
2. If E is an impossible event then $P(E) = 0$
3. If E is a sure event then $P(E) = 1$
4. The sum of probabilities of occurrence and non-occurance of an event E is unity

   i.e., $P(E) + P(\overline{E}) = 1$

Limitations of classical definition of probability

1. This definition does not help if the cases are not equally likely.
2. Enumeration of all equally likely cases are not practicable.
3. This definition fails if the total number of possible cases is infinite.

#Empirical Definition of Probability

If an event E occurs m times in n repetitions of a random experiment then the value approached by the relative frequency $\frac{m}{n}$ when n becomes infinitely large, is said to be the probability of an event E.

Symbolically, $P(E) = \lim_{n \to \infty} \frac{m}{n}$ provided that the limit exists.

Axiomatic Probability

Andrey Nikolaevich Kolmogorov, a Soviet mathematician has introduced the new concept of probability with the help of set theory.

Sample Space

The collection of all possible outcomes of a random experiment is known as a sample space. The sample space is denoted by S. Each outcome of an experiment is known as a sample point. The total number of sample points of a sample space is denoted by $n(S)$.

Favourable cases

The collection of such sample points of a sample space which satisfies the given condition is known as the favourable cases.

Probability

If S is the sample space of a random experiment with number of sample points $n(S)$ and the number of sample points favourable to an event E be $n(E)$, then the probability of happening an event $P(E)$ is defined by

$$P(E) = \frac{n(E)}{n(S)}$$

The probability of happening an event $P(E)$ satisfies the following conditions:

1. $0 \leq P(E) \leq 1$
2. $P(S) = 1$
3. If $E_1, E_2, E_3, ...$ be a sequence of mutually exclusive events of S then

$$P(E_1 \cup E_2 \cup E_3 \cup ...) = P(E_1) + P(E_2) + P(E_3) + ...$$

#Two Basic Laws of Probability

Addition Theorem (Theorem of total probability)

If A and B are two events with their respective probabilities $P(A)$ and $P(B)$ then the probability of occurrence of at least one of these two events denoted by $P(A \cup B)$ is given by

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

where $P(A \cap B)$ is the probability of simultaneous occurrence of the events A and B.

Corollary 1: If A and B are two mutually exclusive events then

$$P(A \cup B) = P(A) + P(B)$$

Corollary 2: If A, B and C are three events, then the probability of occurrence of at least of one them is

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

If the events are mutually exclusive, then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

Multiplication Theorem (Theorem of compound probability)

If two events A and B are independent, then the probability of their simultaneous occurrence $P(A \cap B)$ is equal to the product of their individual probabilities.

Mathematically, $P(A \cap B) = P(A) \cdot P(B)$

Corollary 1: If A, B and C are three independent events, then the probability of their simultaneous occurrence is

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

Corollary 2: The probability that the event A will not happen is $1 - P(A)$. Therefore, the probability that none of the two independent events A and B will happen, is

$$[1 - P(A)][1 - P(B)].$$

$\therefore$ The probability that at least one event will happen is $1 - [1 - P(A)][1 - P(B)]$.

Note: The probability of the simultaneous occurrence of two independent events A and B can also be written as

$P(AB)$ or $P(A \text{ and } B)$.